

Jagiellonian University in Kraków

Faculty of Physics, Astronomy and Applied Computer Science

Mateusz Winiarski

Student ID no.: 1176241

**Sensitivity Comparison of Atomic Interferometry
and Deflectometry for Gravitational
Measurements on Positronium Using Monte
Carlo Simulations**

Master's thesis in physics (path: experimental physics)

Thesis written under the supervision of
dr Sushil Sharma

Marian Smoluchowski Institute of Physics, JU

Kraków, 2026

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Abstract

The weak equivalence principle has not been tested directly on a purely leptonic matter–antimatter system. Positronium, the bound state of an electron and a positron, provides such a system, and its metastable 2^3S state lives long enough to serve as a test particle for inertial sensing. The feasibility of inertial sensing with a 2^3S positronium beam is investigated through Monte Carlo simulations of two proposed schemes: a classical moiré deflectometer and a Mach–Zehnder interferometer. The Monte Carlo model tracks individual positronium atoms, assigning their velocities according to a velocity distribution based on measured 2^3S beam characteristics, an angular divergence in the milliradian range, and lifetimes sampled from an exponential distribution. The acceleration is inferred from the displacement or phase shift of the periodic pattern measured with a third, scanning grating. For a mean velocity of 10^5 m/s and acquisition time of 5.3 weeks the interferometer’s minimum detectable acceleration is (140 ± 1) m/s² (tens of times better than with the deflectometer), in accord with the analytically predicted order. The interferometer sensitivity is optimal at a beam divergence of about 1.6 mrad and remains within a broad, tolerant region of the velocity–divergence plane. These results identify the interferometer as the preferred scheme for a precise gravitational measurement on positronium and quantify the beam conditions required for such measurements.

Keywords: positronium; antimatter gravity; Mach–Zehnder interferometer; moiré deflectometry; Monte Carlo simulations; inertial sensing.

Streszczenie

Słaba zasada równoważności nie została dotąd zbadana bezpośrednio w układzie czysto leptonowym materia–antymateria. Pozytonium, stan związany elektronu i pozytonu, stanowi taki układ, a jego metastabilny stan 2^3S żyje wystarczająco długo, aby służyć jako cząstka testowa w pomiarach inercyjnych. Wykonalność pomiarów inercyjnych z użyciem wiązki pozytonium w stanie 2^3S analizowano za pomocą symulacji Monte Carlo dla dwóch proponowanych schematów: klasycznego deflektometru moiré oraz interferometru Macha–Zehndera. Model Monte Carlo śledzi pojedyncze atomy pozytonium, przypisując im prędkości zgodnie z rozkładem prędkości opartym na zmierzonych charakterystykach wiązki 2^3S , rozbieżność kątową rzędu miliradianów oraz czas życia losowany z rozkładu wykładniczego. Przyspieszenie wyznacza się na podstawie przesunięcia okresowego wzoru bądź jego przesunięcia fazowego, mierzonych za pomocą trzeciej, skanującej siatki. Dla średniej prędkości $10^5 \frac{\text{m}}{\text{s}}$ i czasu akwizycji 5,3 tygodnia minimalne wykrywalne przyspieszenie dla interferometru wynosi $(140 \pm 1) \frac{\text{m}}{\text{s}^2}$ (kilkadziesiąt razy lepsze niż w przypadku deflektometru), zgodnie z przewidywanym analitycznie rzędem wielkości. Czulość interferometru jest największa dla rozbieżności kątowej wiązki około 1,6 mrad i pozostaje wysoka w szerokim, tolerancyjnym obszarze płaszczyzny prędkość–rozbieżność. Wyniki te wskazują interferometr jako preferowany schemat do precyzyjnych pomiarów grawitacyjnych na pozytonium oraz określają ilościowo warunki, jakie musi spełniać wiązka pozytonium do przeprowadzenia takich pomiarów.

Słowa kluczowe: pozytonium; grawitacja antymaterii; interferometr Macha–Zehndera; deflektometria moiré; symulacje Monte Carlo; pomiary inercyjne.

*When I'm left home alone
I know all the explosions from Structure of Matter by heart
Once again, I've broken the entire internet
Suddenly, Nuclear Physics fell into my hands
I'll be calculating all night long

I'll draw nuclei all over my wall
But please, oh please, don't tell the Dean*

– – SMP (Studio of Music and Poetry), 'Nuclei on Wall'
from the upcoming album 'Study Me Half'

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Introduction

1.1 Positronium and its applications in fundamental physics and gravitational studies

Positronium is an electrically neutral bound state of an electron e^- and a positron e^+ , which as a purely leptonic matter–antimatter system is a clean system for experiments involving gravitational interaction with antimatter, in contrast to, e.g., an antihydrogen atom, consisting of an antiproton (antibaryon) and a positron [1]. Unlike antihydrogen, which is a stable particle, positronium decays rapidly in an annihilation process, which makes precise experiments challenging, especially with weak interactions such as gravity.

The lifetime of positronium is determined by the total spin of its components, which dictates the decay mode. In the singlet (*para*) state (1^1S_0 in spectroscopic notation) electron and positron spins are antiparallel, allowing decay into 2γ (or any other even number of photons) with a mean lifetime $\tau = 125$ ps. When the spins of the electron and the positron are parallel, they form *orthopositronium* (1^3S_1 state), which decays to 3γ with a mean lifetime $\tau = 142$ ns.

In order to determine the response of positronium in a field (e.g., gravitational), states with the longest possible lifetimes are preferable. To simply increase the lifetime of orthopositronium, one can excite it to a metastable state. This can be achieved by laser excitation to the 3^3P state, which then de-excites to the 2^3S state with a branching ratio 12% within 35 ns [2]; the process is shown in Figure 1.2. The state 2^3S is metastable (direct deexcitation to the ground state 1^3S is

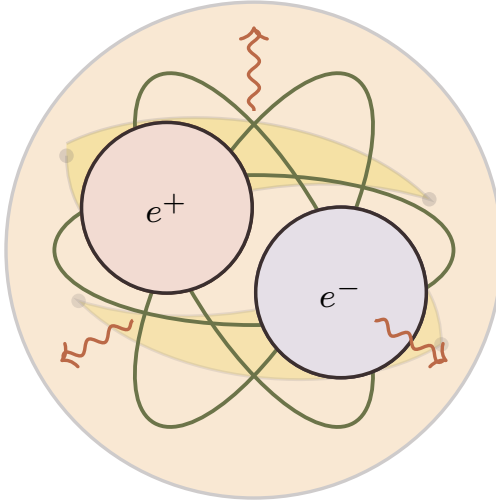


Figure 1.1: Schematic presentation of positronium, an electron–positron bound state. Three gamma photons coming from an orthopositronium decay are also shown.

forbidden due to parity conservation) and decays by self-annihilation to 3γ with a mean lifetime of $\tau = 1.142 \mu\text{s}$.

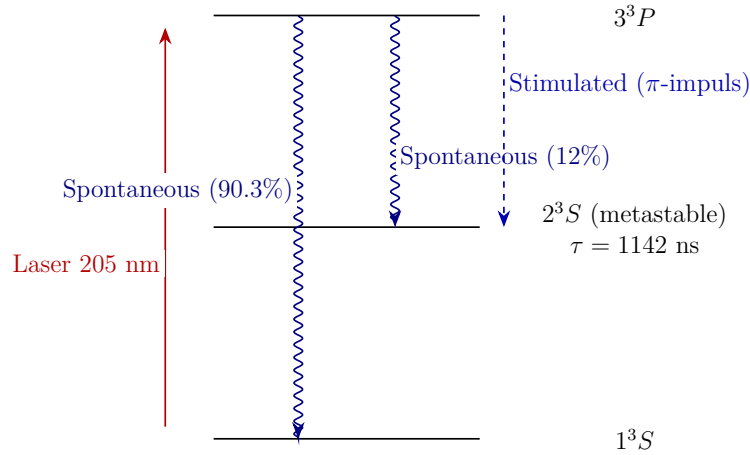


Figure 1.2: Jabłński diagram of metastable 2^3S o-Ps state generation.

It is possible to create even longer-lived states, such as Rydberg states (with mean lifetimes in the range of milliseconds), but such states are more susceptible

to electric field gradients due to their large electric polarisability α ($\alpha \propto n^6$, where n is the principal quantum number) [3, 4].

1.2 Production of a metastable 2^3S positronium beam

In order to measure the gravitational effect on positronium, one must first obtain a beam of metastable positronium atoms. Such a beam can be obtained in a two-step process: converting positrons (e^+) into a positronium cloud, and then selecting positronium atoms with the selected velocities and exciting them as shown in figure 1.2.

Throughout this thesis, the x -axis is aligned with the mean propagation direction of the positronium beam. The y -axis is vertical and defines the sensitive direction, along which gravitational effects are measured through a displacement of the fringe pattern. The z -axis completes a right-handed coordinate system. The gratings are periodic along the y -direction, with their bars aligned along the z -direction.

Positrons are produced via β^+ decay of a ^{22}Na source, and then are slowed down to energies of a few eV using a neon moderator and a Surko trap. Then they are implanted into a silicon converter containing nanochannels with diameters of 7–10 nm, where they diffuse into the SiO_2 walls, combine with electrons and are emitted as Ps atoms into the surrounding vacuum [5].

While positronium ions can be accelerated using techniques like electron photodetachment [6], the lowest kinetic energy achieved so far for a positronium beam has been 1 eV [7]; however, to perform experiments on gravitational interaction, one should ideally obtain a kinetic energy of the atoms of around 50 meV (which will be shown later), which is of the order of room-temperature energies. To obtain such a beam, one can use Doppler selection to choose and excite only positronium atoms with a selected velocity component along the horizontal x axis, and then collimate them using a single slit (particularly because the experimental geometry

is two-dimensional, so the beam can be divergent along the orthogonal horizontal z axis). This method was shown in [8]: after creating a bunch of positronium atoms in vacuum, orthopositronium atoms are excited by a nanosecond ultraviolet laser pulse of a wavelength $\lambda = 205$ nm perpendicular to the propagation direction from the 1^3S state to the 3^3P state. The laser frequency is tuned so that, due to the Doppler effect, only positronium atoms with a specific matching velocity are resonantly excited and survive. By choosing the delay between production and excitation, one can select the velocity of the resulting beam with a precision (velocity spread) of $10^4 \frac{\text{m}}{\text{s}}$. A diagram of this method is shown in Figure 1.3.

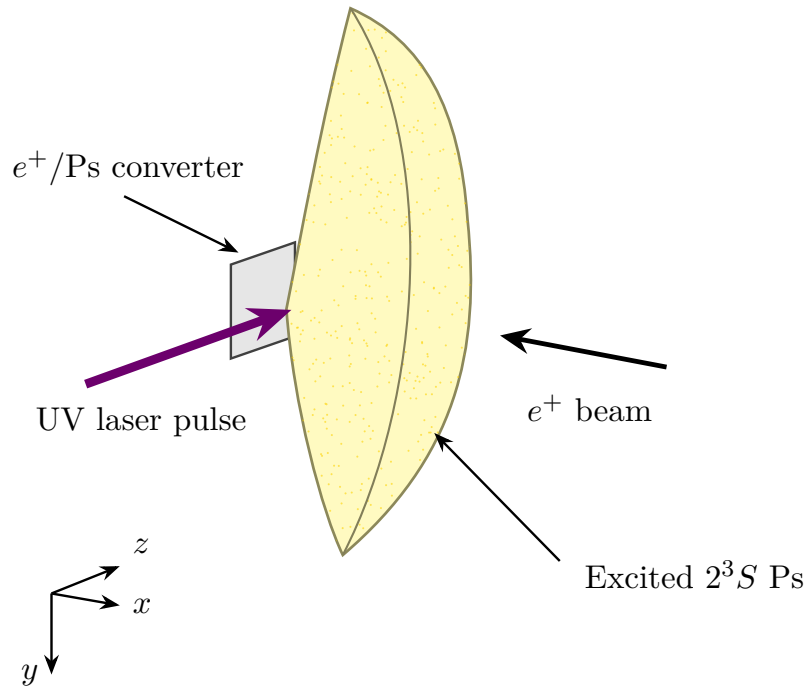


Figure 1.3: Schematic of positronium production and Doppler selection, showing the incident positron beam, the e^+/Ps converter, the UV laser pulse, and the excited 2^3S positronium. Adapted from Mariazzi *et al.* [4]

Table 1.1 lists the efficiencies of the particular steps of this method based on current knowledge.

Recent studies indicate that this method can yield up to 10^5 Ps atoms at a

Process	Efficiency
$e^+ \rightarrow \text{Ps conversion (in room temperature energies)}$	28% [9]
$1^3S \rightarrow 3^3P$ conversion	$\sim 14\%$ [10]
$3^3P \rightarrow 2^3S$ stimulated conversion	$\sim 30\%$ [11]
overall	$\sim 1.18\%$

Table 1.1: Efficiency of $e^+ \rightarrow 2^3S$ positronium conversion steps as of 2026. Assuming a bunch of 10^7 positrons, we can obtain 1.2×10^5 2^3S positronium atoms per bunch.

given horizontal velocity v_h ; after collimation, this gives a flux of $N_a = 123 \theta_{mrad}$, where θ_{mrad} is the angle of divergence in milliradians.

1.3 Multiphoton detector for the detection of positronium annihilation

Positronium is a short-lived bound state and therefore cannot be observed directly. Instead, it is identified through its annihilation products. Depending on its spin state and interaction with the surrounding medium, positronium annihilation results predominantly in two or three photons, whose total energy is approximately $2m_e c^2 = 1022$ keV. Positronium atoms can be used as an excellent probe for both fundamental physics and medical physics applications. In recent years, the J-PET Collaboration has demonstrated that the J-PET system can be used for studies of positronium properties and decays, including positronium imaging and tests of discrete symmetries [12, 13, 14, 15, 16].

A PET scanner consists of rings of detection elements around the region of interest. A conventional PET scanner detects two approximately back-to-back 511 keV annihilation photons in temporal coincidence. The positions of their interactions define a line of response (LOR) along which the annihilation occurred. By combining all coincidence events of a PET scanner, the spatial distribution of positron annihilations can be calculated in three dimensions.

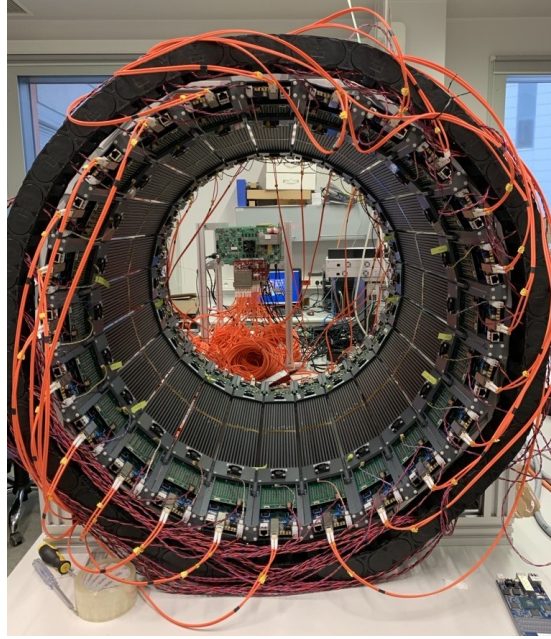


Figure 1.4: Photograph of the Modular J-PET detector in the laboratory. It consists of 24 standalone detector units [16].

However, for the studies of positronium decays, it is of utmost importance that the detector should be able to register events with more than two photons simultaneously. This is especially important for the detection of three photons emitted in the decay of ortho-positronium as well as for the measurements in which the annihilation photons are registered together with additional prompt photons. The Jagiellonian Positron Emission Tomograph (J-PET) detector is made of long plastic scintillator strips (see Fig.1.4). Due to the triggerless data acquisition, it can register multiple photons simultaneously [16]. A feasibility study has been performed with two modular J-PET detectors in a positron beamline. The obtained time-of-flight resolution was about 125 ps (σ) and the annihilation-vertex resolution along the beam direction was about 4 mm (σ) [17]. Importantly, the obtained values enable us to distinguish between annihilation vertices separated by 10–20 mm, which is the requirement of the grating–stopper readout studied in the present thesis.

In the positronium-beam experiments considered in this thesis, such a multi-photon detector can be used to determine where the positronium atom annihilates along the experimental apparatus. In particular, the detector must distinguish annihilations occurring in vacuum from those occurring on the material gratings or on the stopper. The reconstructed annihilation position and photon multiplicity therefore provide the information required for the third-grating detection method introduced in Sec. 2.4 [17]. Modular J-PET detection units have therefore been proposed and experimentally evaluated as position-sensitive detectors for inertial-sensing studies with positronium atoms [17].

It can be easily calculated from the free fall equation of motion that the mean displacement of a positronium atom in Earth's gravitational field over its lifetime lies in the sub-nanometre range, which makes any deflection practically undetectable. A solution to this challenge will be introduced and discussed in section 2.4.



Atomic deflectometry and interferometry for gravitational measurements

2.1 Principles of atomic deflectometry and interferometry

Atomic interferometry is an area of physics that utilises the phenomenon of wave interference (superposition of coherent waves) using matter waves. Although matter-wave interference had been demonstrated earlier, modern separated-beam atom interferometers were realized in the early 1990s, notably the three-grating sodium interferometer of Keith *et al.* [18]. Since its experimental confirmation, atomic interferometry has been widely applied in fundamental physics experiments where high sensitivity is needed, such as gravimetry [19], tests of relativistic theories (such as equivalence principle tests), measurements of fundamental constants (α , G and $\hbar$) and high-precision inertial sensing (with applications in space probes).

The systems considered in this thesis show two distinct approaches: whereas Mach–Zehnder interferometry is based on quantum matter-waves interferometry, moiré deflectometry is a fully classical method based on particle trajectories.

2.2 Moiré deflectometry

A moiré (from French: shimmering, rippled) pattern is a pattern produced by the light shadow when two or more overlaid repeating opaque patterns (e.g. grids or parallel lines) form an optical illusion of a new, larger repeating pattern. The effect arises when the two patterns are slightly rotated relative to each other, or when the spacing of one of the patterns is slightly different from that of the other, causing the lines to periodically fall in and out of alignment. An example of such a pattern is shown in Figure 2.1.

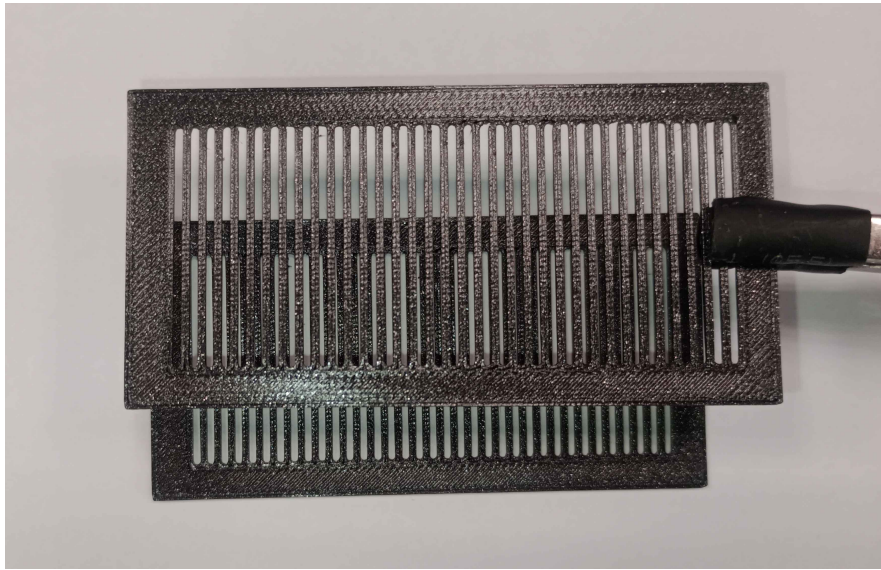


Figure 2.1: A moiré pattern, created with 3D-printed elements. Both gratings have the same period $d = 2$ mm and opening width $a = 1$ mm.

Such an effect can be used in high-precision experiments, due to its high sensitivity to small changes in the displacement of the patterns. The effect can also be observed with particles instead of light: in this case, with stationary gratings, the position of the moiré pattern is affected by a change in the path of motion of the particles. When the particles pass through the first grating, they create a periodic intensity distribution, which then falls on the second grating. The displacement of the distribution relative to the second grating produces a

displacement of the moiré pattern. The measurement of this displacement allows determining the deflection angle of the trajectories, and hence the acting force.

In fundamental physics, moiré patterns are employed in deflectometers: highly sensitive instruments used to measure small forces that curve the particles' trajectories. A typical deflectometer consists of a collimated particle beam, a pair of gratings with an equal grating period d and an equal grating width a (hence the same open fraction $f = a/d$), and a screen. (Alternatively, a setup with three gratings can be used; in this configuration, the first grating acts as a source of spatial coherence, so the original beam does not need to be collimated.) The distances between the particle source and the first grating, between the gratings, and between the second grating and the screen are equal and denoted by L . The particle beam, while passing through the gratings, is blocked when it meets the opaque regions of the gratings, and the transmitted part of the beam produces a moiré pattern with a period d on the screen. The resulting pattern can be compared with a reference pattern of light passing through the same gratings (since light is not deflected by the force on a laboratory scale), and the relative shift between the two patterns yields the deflection. Alternatively, one can reverse the direction of the force (e.g. by rotating the apparatus by 180°) to observe a pattern shifted in the opposite direction, but of the same magnitude. This principle is illustrated in Figure 2.2.

A moiré deflectometer was proposed by the AEgIS collaboration to test the gravitational interaction of antiprotons, and experiments were conducted in 2014 [20]. In a proof-of-concept experiment conducted at CERN's Antiproton Decelerator, a moiré pattern was obtained and a microscopic relative shift between the antiproton moiré pattern and the Talbot–Lau light pattern was observed. A moiré deflectometer experiment for antihydrogen is under preparation by the AEgIS collaboration as of 2025 [21].

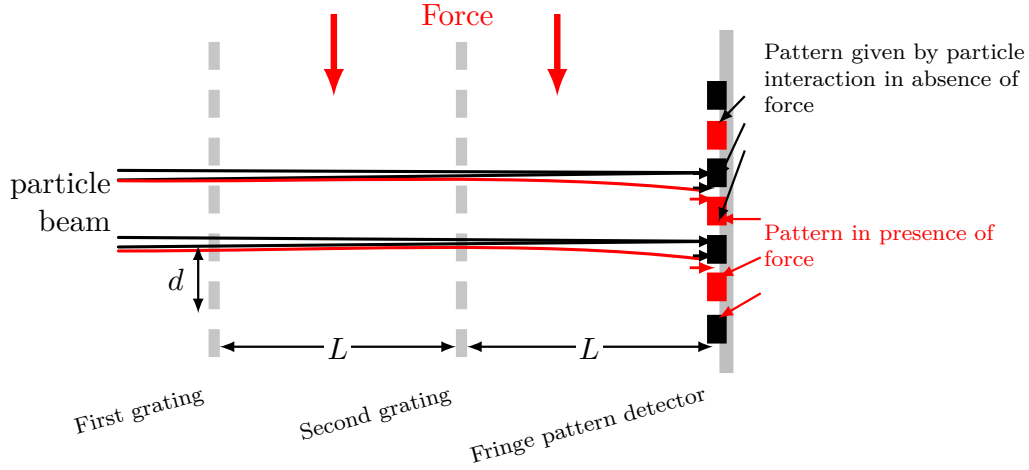


Figure 2.2: The principle of operation of a two-grating moiré deflectometer. In the absence of a force, particles form a black pattern on a screen; in the presence of a force, they form a downward-shifted red image. The vertical and horizontal axes are not shown to scale. Adapted from Mariazzi *et al.* [4].

2.3 Mach–Zehnder interferometry

A Mach–Zehnder interferometer is an optical device used to measure the relative phase difference between two light beams. The device was originally created by Ludwig Zehnder in 1891 and refined by Ludwig Mach in 1892 [22]. The Mach–Zehnder geometry was later extended from optical to matter waves, first in atom interferometers based on diffraction gratings and subsequently in electronic interferometers in mesoscopic systems [23, 24].

The idea behind the interferometer is to divide the original beam by a beam splitter into two beams: the test beam and the reference beam, each beam is then reflected by a mirror and both beams are recombined in the second beam splitter, as shown in Figure 2.3. To construct such an interferometer using matter waves, one can replace optical beam splitters with diffraction gratings, laser beams (via Raman or Bragg transitions) or magnetic mirrors.

The scheme of the Mach–Zehnder interferometer in the context of positronium gravity measurements is shown in Figure 2.4. Its main benefit in comparison with

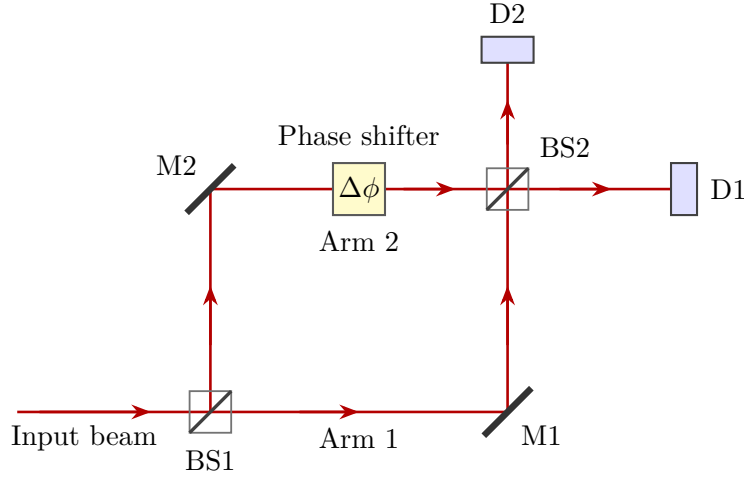


Figure 2.3: Schematic of an optical Mach-Zehnder interferometer. The input beam is divided at beam splitter BS1, reflected by mirrors M1 and M2, and recombined at beam splitter BS2. A phase shifter introduces a relative phase shift $\Delta\phi$ between the arms; detectors D1 and D2 record the two output beams.

the moiré deflectometer is the possibility of using laser standing waves as gratings, which offers two advantages: no particles are lost by annihilation in the gratings, and the grating period can be reduced by about two orders of magnitude.

A standing light wave with wave vector k produces a spatially periodic intensity, $I(y) \propto \cos^2(ky)$, and hence a periodic optical-dipole potential acting on the atom; this potential is what plays the role of a diffraction grating. By tuning the depth and the interaction time of the potential, the fraction of atoms transferred between the coupled momentum states can be controlled, so that the standing wave acts as a coherent beam splitter or mirror for the matter wave [25, 26].

Instead of using a physical grating to deflect a positronium atoms beam [27], the beam is diffracted by a standing laser wave acting as an optical grating, which is known as the Kapitza-Dirac effect [28]. In this effect, an atom passing through the grating experiences a momentum difference of $\Delta p = 2n\hbar k$, where n is a natural number. The laser beam has to be wide enough, i. e. wider than the Talbot length $L_T = \frac{2d^2}{\lambda_{dB}}$ (where d is the grating period, and $\lambda_{dB} = h/mv$ is de Broglie wavelength of the particle), for the effect to operate in the Bragg regime, so the

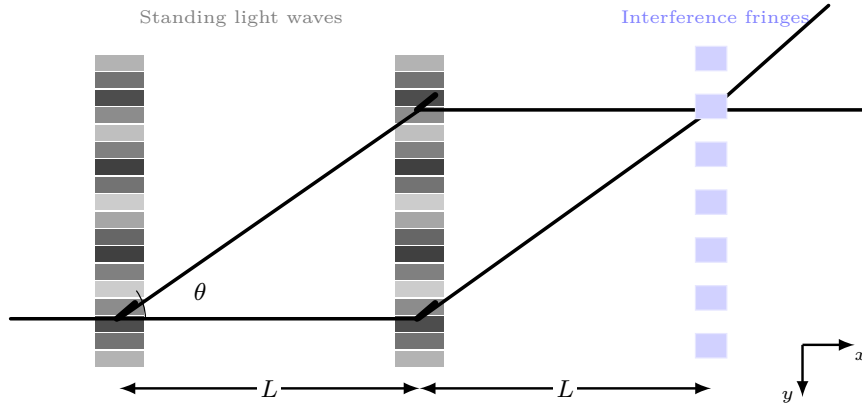


Figure 2.4: Scheme of a positronium atom beam Mach–Zehnder interferometer, in a configuration with two standing-wave light gratings and an interference fringe screen/detector. Adapted from Mariazzi *et al.* [4].

grating produces only one diffraction peak and leaves the original beam [25] (i.e. two possible momentum states $n = 0$ and $n = 1$ are possible). The wavefunction of an atom becomes a coherent superposition of two momentum states:

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left(|p\rangle + e^{i\phi} |p + 2\hbar k\rangle \right). \quad (2.1)$$

Since the two beams have different momenta, they propagate in different directions, thus creating two arms of the interferometer.

In this scheme, as in most quantum Mach–Zehnder interferometers, the optical phase is replaced by the quantum phase evolution of the particle beam. In this case, the phase change does not arise from any physical obstruction on the way of the beam, but from a change of potential, which accumulates over time as a change of action S , thereby changing the phase ϕ :

$$\phi(t) = \frac{S(t)}{\hbar} = \frac{1}{\hbar} \int_0^t [E_k(t') - E_p(t')] dt'. \quad (2.2)$$

For a uniform acceleration a along the sensitive direction, the phase shift can be derived in a freely falling frame. Let $t = L/v$ be the flight time between

successive grating planes, so that the total time from the first grating to the detector plane is $2t$. In this frame, the apparatus accelerates at $-a$, giving the following acceleration-induced displacements at times 0, t , and $2t$:

$$\delta y_1 = 0, \quad \delta y_2 = -\frac{1}{2}at^2, \quad \delta y_3 = -\frac{1}{2}a(2t)^2 = -2at^2. \quad (2.3)$$

The fringe phase depends on the combination $\delta y_1 - 2\delta y_2 + \delta y_3$; the factor of two arises from the opposite diffraction of the two paths at the second grating [23]. Choosing the phase convention in which positive a gives positive $\Delta\phi$, we obtain

$$\begin{aligned} \Delta\phi &= -\frac{2\pi}{d}(\delta y_1 - 2\delta y_2 + \delta y_3) = -\frac{2\pi}{d}(at^2 - 2at^2) \\ &= \frac{2\pi at^2}{d}. \end{aligned} \quad (2.4)$$

The resulting fringe pattern seen on the screen in this scenario is a sinusoidal pattern, with a period equal to the grating period; since the grating is formed by a standing wave, this period is $\lambda/2$.

In a Mach–Zehnder interferometer (and any other interferometer), to obtain clear and distinct interference fringes, the input beam has to be coherent, which is a disadvantage of this method. In a real experiment, the beam of positronium atoms must be collimated in both velocity and angle of incidence, which limits the number of atoms that can be used in the experiment.

2.4 Third-grating detection scheme

Despite proposals of atomic deflectometry/interferometry, direct detection of the gravitational deflection is not straightforward. The main difficulty is the short interrogation time with the gravitational field, limited by the mean positronium lifetime – given Earth’s gravitational acceleration and the mean lifetime of 2^3S positronium, the resulting beam displacement does not exceed one nanometre. To overcome this limitation, a setup including a third grating and a stopper is introduced, as proposed in [4] and as shown in Figure 2.5. By scanning the third

grating along the y -axis and counting the positronium atoms reaching the stopper, a periodic transmission curve is obtained. Gravitational deflection shifts this curve relative to the reference measurement, converting the sub-nanometre beam displacement into a measurable shift of count modulation.

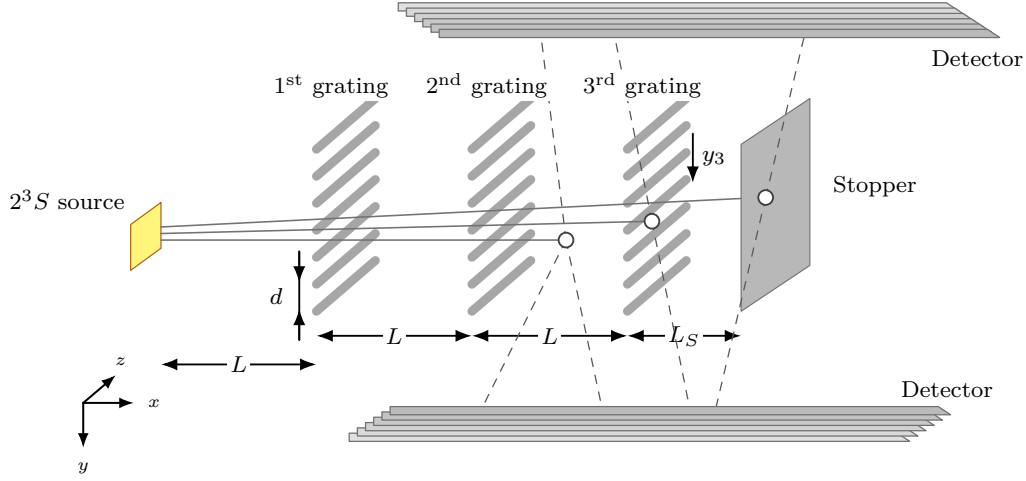


Figure 2.5: Scheme of the third-grating detection method. Positronium atoms passing through the gratings are stopped by the stopper, while photons (dashed lines) from their annihilation (circles) are registered by the surrounding detectors. The third grating can be moved along the y -axis with nanometre-scale precision, as indicated by the y_3 arrow. The grating period and the distance between the third grating and the stopper are not shown to scale. Adapted from Mariazzi *et al.* [4].

To the two-grating scheme a third grating is added, with the same grating period d and open fraction f as the first two gratings, at a distance L from the second grating. Behind the third grating a solid stopper is placed. The third grating and the stopper should be made from solid (non-transparent to the beam) materials, regardless of the transparency of the remaining ones (i.e. if they are solid or made of a standing-wave laser beam). The distance between them (denoted on the figure as L_S) should be sufficiently large for the external detector to distinguish between positronium atoms annihilated on the third grating and the stopper by its x -coordinate. In the present design, this distance is set to 2 cm.

In this scheme, each measurement is performed for a different position y_3 of the

third grating along the y -axis, which, using commercially available piezoelectric nanopositioning systems, can be moved with a positioning resolution down to 0.05 nm [29]. Thanks to the resolution, the entire length of a grating period d can be scanned in nanometric steps, yielding from several hundred up to several thousand measurement points, even for $d = \lambda/2 = 656$ nm in the Mach–Zehnder interferometer. For each position, the number of positronium atoms annihilated on the third grating, on the stopper, and in the space between them is collected.

Note that this method distinguishes between positronium atoms decaying ‘mid-air’ and decaying on a physical grating or a stopper: whereas in vacuum an orthopositronium atom decays into three gamma photons due to spin conservation, in matter the positron from an orthopositronium atom can pick off an electron with opposite spin and annihilate into two gamma photons with a probability over 98% in a few nanoseconds [30]. There are currently no precise experimental data about the incidence of the pick-off effect in matter for the 2^3S state, although it has been suggested in [31] that the pick-off effect is even stronger in the 2^3S state than in the 1^3S (ground) state, due to larger size of an atom, longer lifetime and smaller binding energy.



Monte Carlo modeling and sensitivity analysis

This chapter presents the results of Monte Carlo simulations, comparing the sensitivities of the previously described methods, and compares the simulation results with the analytical estimates presented in the article by Mariazzi *et al.* [4].

The two schemes are modelled at different levels. For the moiré deflectometer the simulation propagates individual positronium atoms along classical trajectories through material gratings. For the Mach–Zehnder interferometer the atom–light interaction and the coherent evolution through the optical gratings are not solved from first principles. Instead, the interference probability expected from theory is used as the density from which individual detection events are sampled. The Mach–Zehnder calculation is therefore an event-level, phenomenological Monte Carlo model of the interferometer readout, incorporating the velocity distribution, angular acceptance, finite lifetime, and counting statistics of the beam.

3.1 Monte Carlo model of the moiré deflectometer

The simulations presented in this section follow the motion and annihilation of positronium atoms through an experimental setup described in Section 2.2. Scanning the position of the third grating over one grating period yields the resulting moiré pattern.

3.1.1 Classical trajectory simulation

In simulations, we use as reference the geometry setup described in Sec. 2.4 and Ref. [4]. The distance L between two grating planes was fixed to 20 cm, and the distance L_S from the third grating to the stopper to 2 cm. The grating period d and the open fraction f of the moiré deflectometer are set to the values 40 μm and $1/3$, respectively. The third material grating was modeled as a grating of open fraction $f = 1/3$. The lifetime τ of the 2^3S metastable state of the positronium is set to $\tau = 1.142 \mu\text{s}$ [2]. The mean beam velocity v_0 and the angular divergence θ_{max} are varied in the simulations to study the sensitivity of the setup.

In each run of the simulation, N_b bunches of S positronium atoms were simulated, and each of the $N = N_b S$ atoms was assigned an initial velocity (chosen randomly from a normal distribution with $\mu = v_0$, $\sigma = 10^4 \frac{\text{m}}{\text{s}}$; here, v_0 denotes the mean initial velocity of the positronium beam), an initial angle (chosen randomly from a continuous uniform distribution with bounds $-\theta_{\text{max}}/2$ and $\theta_{\text{max}}/2$, θ_{max} denotes the angular beam divergence) and a lifetime (drawn from an exponential distribution). In the notation used here, θ_{max} is the full angular width of the uniform incidence distribution, $[-\theta_{\text{max}}/2, +\theta_{\text{max}}/2]$, not the maximum absolute angle.

For each particle, its trajectory is determined, assuming parabolic projectile motion in a gravitational field with acceleration a . Then it is determined whether the positronium atom decays in vacuum or is annihilated in one of the gratings or in the stopper. The x -coordinate of the annihilation point is then stored, and the number of atoms annihilated at each grating is determined. The procedure is illustrated in Figure 3.1.

The number of equally spaced positions of the third grating sampled over one grating period d is denoted by N_{off} . The procedure is repeated independently at each position, with a step of d/N_{off} , and the atom count for each position is saved. Figure 3.2 shows an example of the number of annihilations on the third grating, on the stopper, and in the space between them as a function of the third-grating position. The pattern shown in the figure is periodic with a period of d ; the plot

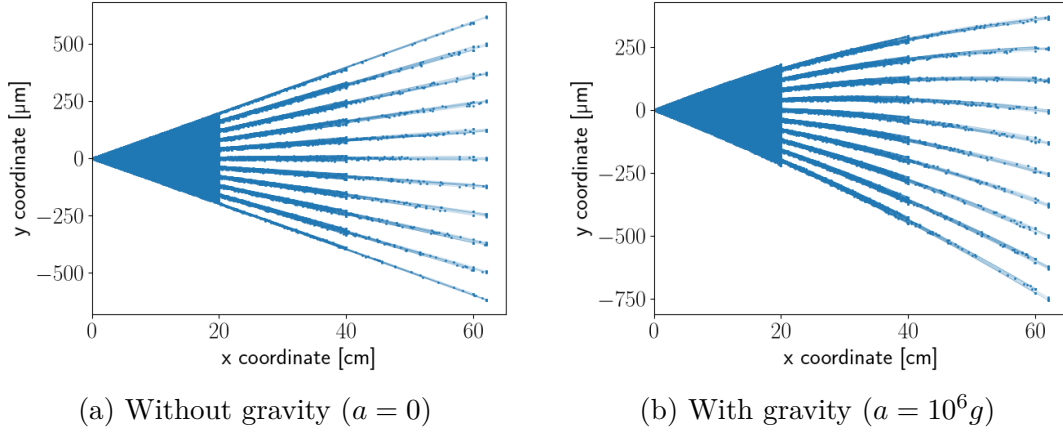


Figure 3.1: Illustration of the tracking of individual positronium atoms (shown as blue points) for the simulation parameters: $v = 10^5 \frac{\text{m}}{\text{s}}$, $N = 40\,000$, $\theta = 2 \text{ mrad}$, $L = 20 \text{ cm}$. Each atom is tracked, its annihilation position is determined, and then its position relative to the gratings is stored (the gratings can be clearly seen in the plots as clusters of the annihilation points at $x = 20, 40, 60 \text{ cm}$). In panel 3.1b, the gravitational acceleration was increased to $10^6 g$ to make the deflection clearly visible. Note that x -axis is in centimeters, while y -axis is in micrometers.

shows only one period of the pattern.

The shift of the moiré pattern in the presence of a force is shown in Figure 3.3. Note that the accelerations used in this simulation were multiples of Earth's gravitational acceleration: since the minimum detectable acceleration by this method is given in [4] as

$$a_{\min} = \frac{d}{2\pi V t^2 \sqrt{N_{\det}}}, \quad (3.1)$$

where V is the visibility of the fringes, given by

$$V = \frac{n_{\max} - n_{\min}}{n_{\max} + n_{\min}}, \quad (3.2)$$

$n_{\min}$ and $n_{\max}$ are the minimum and maximum values on the raw (unfitted) particle count on the stopper $n(y_3)$ respectively, $t = L/v$ is the interrogation time, and $N_{\det}$ is the number of detected atoms per scan, one can estimate that the detectability of the acceleration-based pattern shift scales with $1/\sqrt{N_{in}}$, where N_{in}

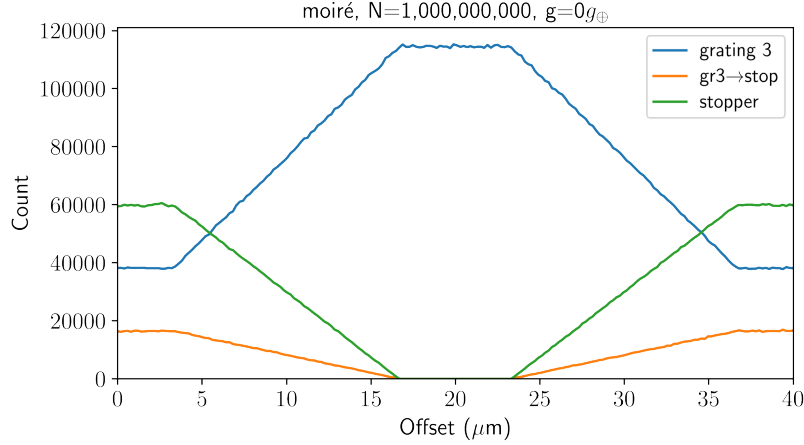


Figure 3.2: A plot of the annihilation count on the third grating (blue), on the stopper (green), and in the space between them (yellow) as a function of the position of the third grating y_3 , for the simulation parameters: $v = 5 \times 10^4 \frac{\text{m}}{\text{s}}$, $N = 10^9$, $\theta = 5 \text{ mrad}$, $L = 20 \text{ cm}$. The count of the particles decaying or annihilating in other places is not shown, because they are outside the plotted range for the given parameters and are not physically meaningful (do not form a pattern). Approximately one in 8700 positronium atoms generated at the input reached the third grating.

is the number of positronium atoms on input. In the case presented in the figure, it can be estimated that one can detect an acceleration of $1g$ with this precision using 10,000 times as many particles.

Figure 3.4 shows the dependence of the minimum detectable acceleration $a_{\min}$ on the angular beam divergence $\theta_{\max}$, given the initial velocity $v_0 = 10^5 \frac{\text{m}}{\text{s}}$. The sensitivity of the deflectometer increases with $\theta_{\max}$, because the number of positronium atoms increases with the collimation angle (the positronium source emits atoms isotropically, and then the slit cuts out the uncollimated part of the beam). In the case of the moiré deflectometer, there is no global minimum of the minimum detectable acceleration, although there are limitations concerning the maximum reasonable beam divergence. Assuming a maximum angular divergence based on the article [4] – $\theta_{\max} = 17 \text{ mrad}$ – the minimum achievable detectable acceleration is equal to $(3408 \pm 31) \frac{\text{m}}{\text{s}^2}$.

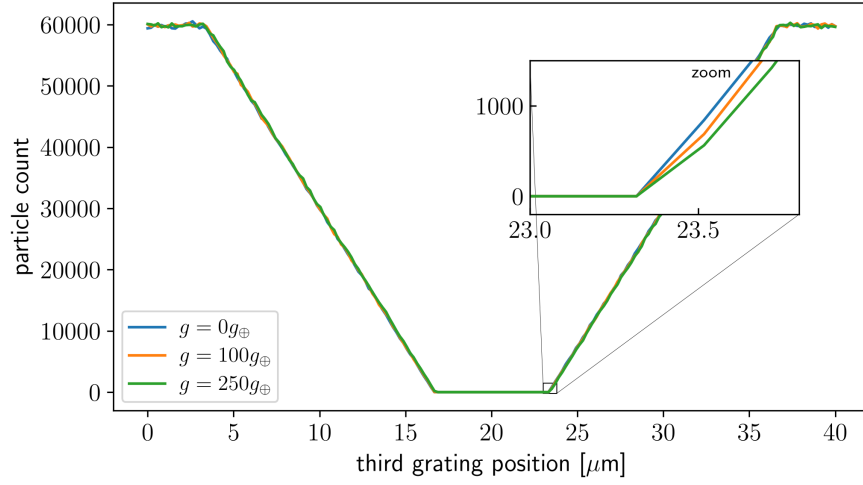


Figure 3.3: The particle count on the stopper as a function of the third-grating position for gravitational accelerations of 0, $100g$, and $250g$, where g denotes Earth's gravitational acceleration. The inset shows an enlarged view of the rising edge near the minimum, highlighting the differences between the three curves.

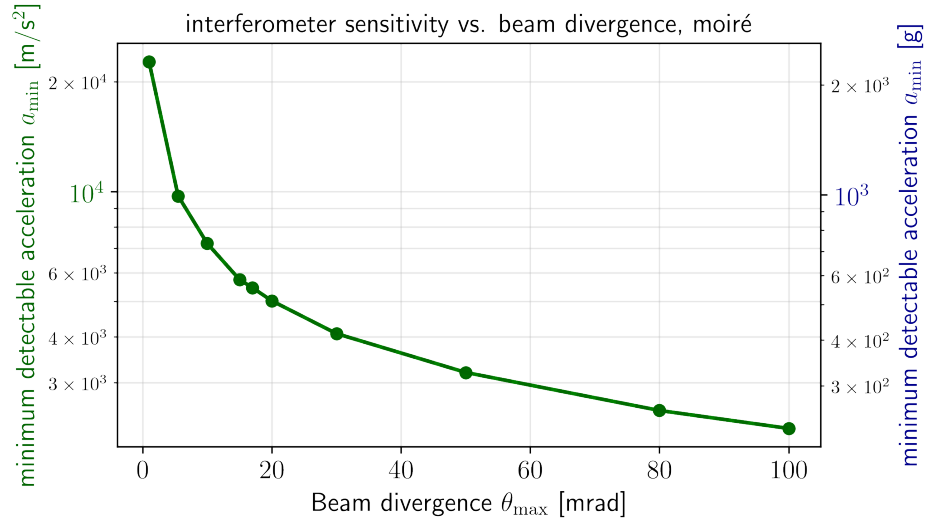


Figure 3.4: The minimum detectable acceleration a_{min} of the moiré deflectometer as a function of the beam divergence θ_{max} . The left axis gives the acceleration in $\frac{\text{m}}{\text{s}^2}$, while the right axis expresses it in units of Earth's gravitational acceleration g .

Figure 3.5 shows a heatmap of the dependence of $a_{\min}$ on the angular beam divergence $\theta_{\max}$ (on the x -axis) and the mean initial velocity v_0 (on the y -axis). Given the constraints on angular divergence described in the previous paragraph, to optimise the function realistically, one can find a minimum for a fixed $\theta_{\max}$. However, the situation is different for the initial velocity v_0 : the minimum is a trade-off between two effects – faster atoms accumulate a smaller gravitational shift, thus the smaller the fringe shift, but also the less time it spends inside the apparatus, and thus the greater the survival probability of positronium atoms. Under these conditions, the lowest achievable value of $a_{\min}$ for a given number of input particles is $a_{\min} = (3382 \pm 30) \frac{\text{m}}{\text{s}^2}$ or $(344.9 \pm 3.1) g$, for $v_0 = 0.9 \times 10^5 \frac{\text{m}}{\text{s}}$. The reported value is the mean of ten independent simulation runs, and the quoted uncertainty is the standard deviation of the results across these runs. This uncertainty quantifies only the statistical scatter between independent Monte Carlo runs. It does not include systematic contributions from grating alignment, beam and source properties, or detector response.

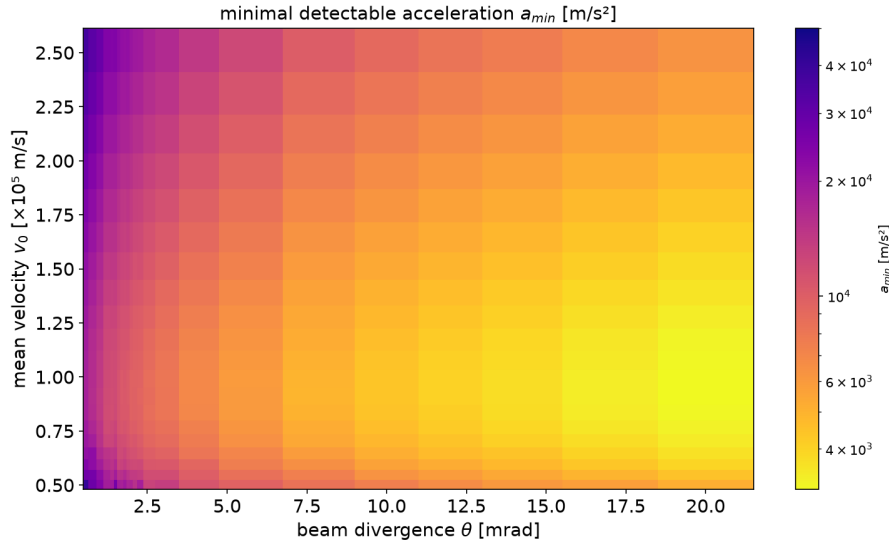


Figure 3.5: The minimum detectable acceleration $a_{\min}$ of the moiré deflectometer as a function of the beam divergence $\theta_{\max}$ and the mean velocity v_0 . The colour scale indicates $a_{\min}$ in $\frac{\text{m}}{\text{s}^2}$.

Knowing that sensitivity scales as $1/\sqrt{N}$, to increase the sensitivity to g , the experiment would have to run $(a_{\min}/g)^2 \approx 119\,000$ times longer than in the simulated experiment (the number of shots $N_s = 400$).

3.2 Event-level Monte Carlo model of the Mach–Zehnder interferometer

In the case of simulating the Mach–Zehnder interferometer, the main idea – simulating a bunch of atoms, tracking each of them and saving its point of annihilation – is the same as in the case of the moiré deflectometer. Also in this case, for each particle an initial velocity, an angle of incidence, and a lifetime are randomly assigned, as described in the previous section. In the underlying Mach–Zehnder interferometer, the positronium state evolves as a coherent superposition of the two paths. During the flight between the first and second gratings, the total relative phase ϕ_{total} accumulates, depending on the positions of the gratings and gravitational acceleration. In this simulation, the total gravitational phase shift during the flight is given by

$$\phi = \frac{2\pi g t^2}{d}, \quad (3.3)$$

where $t=L/v$ is the time of flight (interrogation time), and a phase offset $\phi_0 = 0$ is assumed (in a real experiment it is unknown, but constant, so this offset will be removed later). Then, the interference phase θ at the third, material grating is randomly chosen from $[0, 2\pi)$ using rejection sampling with a probability density function determined by the Born rule (the probability that an atom will be found at a given vertical position):

$$\rho(\theta) = \frac{1}{2\pi} (1 + C_{max} \cos(\theta + \phi)). \quad (3.4)$$

Here C_{max} denotes the assumed intrinsic contrast of the interference pattern prior to the third grating’s readout. We assume $C_{max} = 0.8$ in the present, purely phenomenological model. The Monte Carlo procedure only samples individual

detection events from this density and propagates the effects of a finite lifetime, of a velocity spread, of an angular acceptance and of counting statistics to the reconstructed acceleration.

When the atom has its assigned interference phase θ , the algorithm simulates the action of the third grating: a periodic mask made of alternating bars and gaps with a period d and an open fraction $f = 0.5$. Its positional phase is defined as

$$\phi_y = \frac{2\pi y_3}{d} \mod 2\pi. \quad (3.5)$$

The algorithm then calculates the phase difference to determine whether an atom passes through the grating:

$$\begin{cases} (\theta - \phi_y) \mod 2\pi < 2\pi f & \text{atom passes} \\ (\theta - \phi_y) \mod 2\pi \geq 2\pi f & \text{atom does not pass.} \end{cases} \quad (3.6)$$

If the atom passes through the grating, its remaining lifetime is randomly chosen again (due to the memorylessness of the exponential distribution), and if the remaining lifetime is shorter than $t_s = L_s/v$, the positronium atom annihilates in the gap between the third grating and the stopper; otherwise, it annihilates on the stopper. If the atom does not pass, its annihilation on the third grating is recorded.

The entire procedure above is repeated for each atom at a given third grating position y_3 , and the total count of atoms annihilated on the third grating, annihilated on the stopper, or decaying in between is recorded. Then, the procedure is repeated for all positions in the distance of one period d . Example simulation results are shown in Figure 3.6.

The phase shift from Figure 3.7 can be determined quantitatively by fitting the following equation to the data:

$$N(y_3) = N_0[1 + C \cos(2\pi y_3/d + \phi)]. \quad (3.7)$$

Here C denotes the fitted modulation amplitude of the third-grating transmission

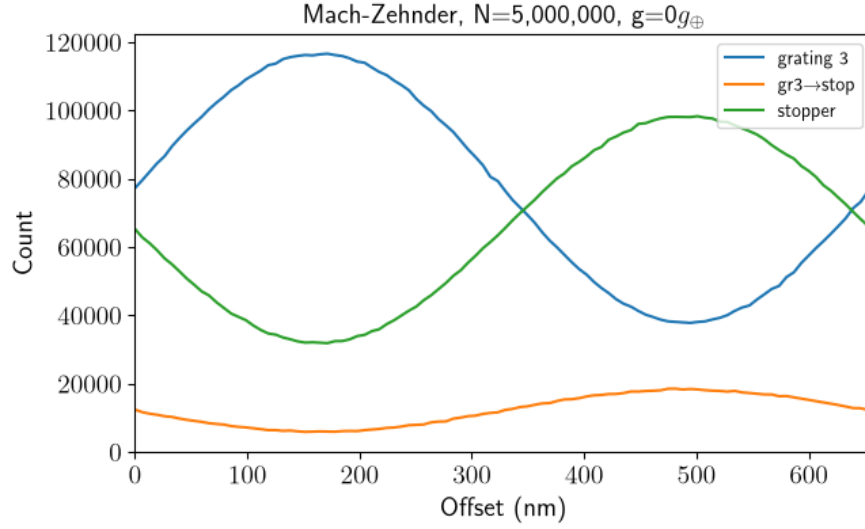


Figure 3.6: Simulated annihilation counts on the third grating (blue), on the stopper (green), and in the gap between them (orange) as a function of the third-grating position y_3 in the Mach–Zehnder interferometer, for $N = 5 \times 10^6$ and zero gravitational acceleration.

curve and should be distinguished from the input intrinsic contrast $C_{\max}$ in Eq. (3.4); V in Eq. (3.2) denotes the visibility evaluated directly from the maximum and minimum stopper counts. From the fitted phase ϕ the gravitational acceleration g can be reconstructed using

$$g = \frac{\phi d}{2\pi \langle t^2 \rangle}, \quad (3.8)$$

where $\langle t^2 \rangle$ is the effective mean square time of interrogation (which is needed because the velocity is not the same for each positronium atom, and atoms with a higher velocity have a higher probability of survival).

To compare the results of the simulation with previous calculations from the literature, we have to calculate the minimum detectable acceleration for 400 shots, i.e. 272 000 positronium atoms per point on the interferometer input. To obtain $a_{\min}$, we have to use Eq. (3.1) also in this case.

Figure 3.9 shows $a_{\min}$ as a function of the angular beam divergence $\theta_{\max}$ for a fixed initial velocity $v_0 = 10^5 \frac{\text{m}}{\text{s}}$. In contrast to the case of the moiré deflectometer

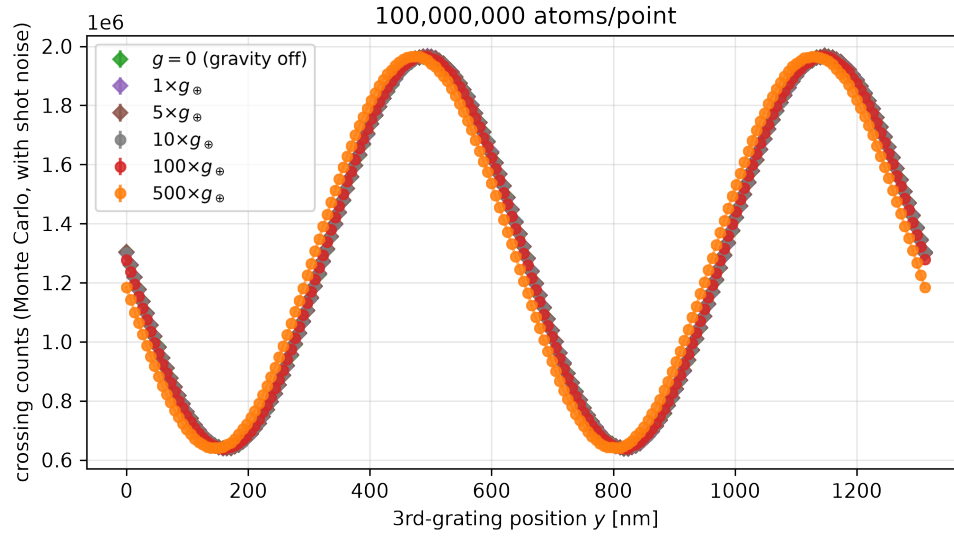


Figure 3.7: Simulated counts of atoms passing through the third grating as a function of the position of the third grating y_3 for accelerations of 0 , g , $5g$, $10g$, $100g$, and $500g$, where g denotes Earth's gravitational acceleration. Each scan point uses 10^8 atoms.

(Figure 3.4), this plot shows a minimum in $a_{\min}$ – for a beam divergence of $\theta_{\max} \approx 1.6$ mrad, $a_{\min} = (140 \pm 1) \frac{\text{m}}{\text{s}^2}$ (averaged over 10 runs). This is due to the trade-off between two effects: an increase in the number of atoms and a decrease in visibility. On the one hand, as the beam divergence increases, the number of positronium atoms increases ($N_0 \propto \theta_{\max}$). On the other hand, the Mach–Zehnder interferometer is very sensitive to angular coherence of the beam – optical gratings accept only atoms with a finite acceptance angle [26], and in the simulation the acceptance is modelled using a Gaussian approximation

$$c_{\text{atom}} \propto \exp\left(\frac{-\theta_{\text{in}}^2}{2\sigma_{\text{acc}}^2}\right), \quad (3.9)$$

where θ_{in} is the angle of incidence and σ_{acc} is a phenomenological width parameter describing the finite angular acceptance in the present simulation. For a standing-

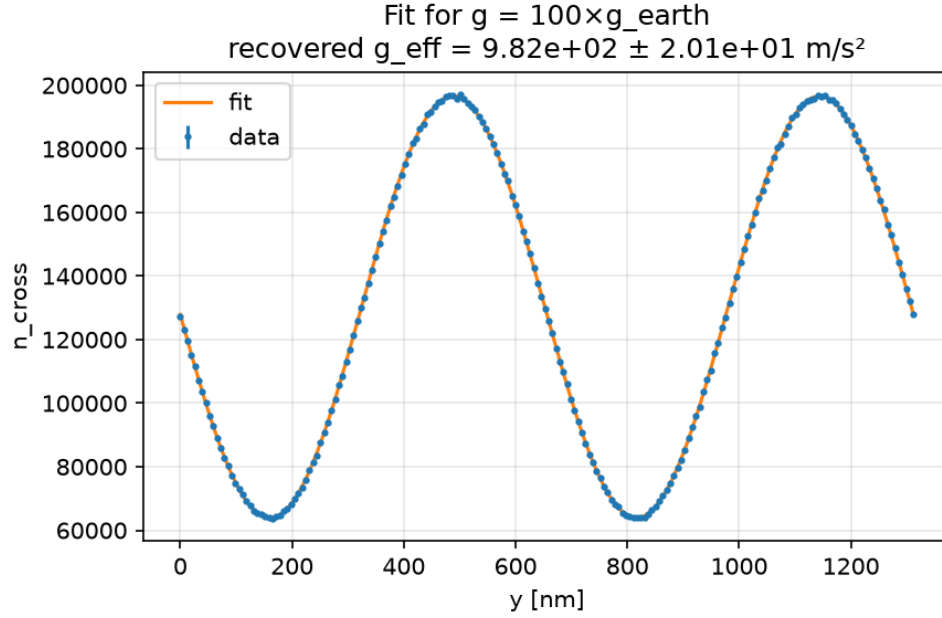


Figure 3.8: Sinusoidal fit (orange line) to the simulated counts of atoms passing through the third grating (blue points) as a function of its position y_3 , for an applied acceleration of $100g$. The acceleration reconstructed from the fitted phase is $(982 \pm 20.1) \frac{\text{m}}{\text{s}^2}$, corresponding to $(100.1 \pm 2.1)g$, where $g = 9.81 \frac{\text{m}}{\text{s}^2}$. Each point on the plot corresponds to 10^8 positronium atoms generated at the input of the simulation.

wave wavelength of $\lambda = 1312 \text{ nm}$, Ref. [4] obtains the diffraction angle

$$2\theta_B = \frac{\lambda_{\text{dB}}}{d} \simeq 5.5 \text{ mrad},$$

from Eq. (6), and uses this angle as a characteristic scale for the required angular collimation of the positronium beam. The value $\sigma_{\text{acc}} = 5.5 \text{ mrad}$ is therefore adopted here as the width of the Gaussian acceptance model. This Gaussian form should not be interpreted as the exact Bragg-reflection probability or as an expression derived in Ref. [4]. The actual angular dependence of Bragg reflection can be more complex within the finite acceptance region [26].

Figure 3.10 shows a heatmap of the minimum detectable acceleration as a

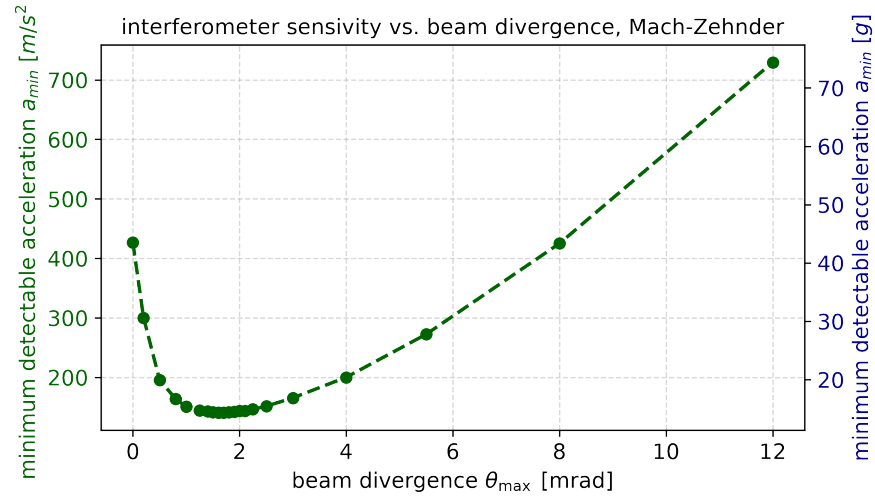


Figure 3.9: The minimum detectable acceleration $a_{\min}$ of the Mach–Zehnder interferometer as a function of the beam divergence $\theta_{\max}$. The left axis gives the acceleration in $\frac{\text{m}}{\text{s}^2}$, while the right axis expresses it in units of Earth’s gravitational acceleration g .

function of the angular beam divergence $\theta_{\max}$ and the mean initial velocity v_0 , showing a minimum at $\theta_{\max} \approx 1.6$ mrad and $v_0 \approx 0.8 \times 10^5 \frac{\text{m}}{\text{s}}$. For these parameters, the minimum detectable acceleration is $a_{\min} = (134.6 \pm 1.2) \frac{\text{m}}{\text{s}^2}$. The optimal region on the velocity-divergence plane is broad, so the experiment is highly tolerant of the exact choice of initial velocity and angular collimation – these parameters can deviate by tens of per cent without significantly degrading the sensitivity.

3.3 Comparative sensitivity analysis

Table 3.1 summarises the sensitivity estimates for both methods: moiré deflectometry and Mach–Zehnder interferometry. For the simulated configurations with a fixed mean initial velocity $v_0 = 10^5 \frac{\text{m}}{\text{s}}$, the Mach–Zehnder interferometer yields a minimum detectable acceleration lower by a factor of approximately 24 than that of the moiré deflectometer, which corresponds to a 593-fold shorter time of experiment, for the same detectable acceleration (since $N \propto 1/a_{\min}^2$).

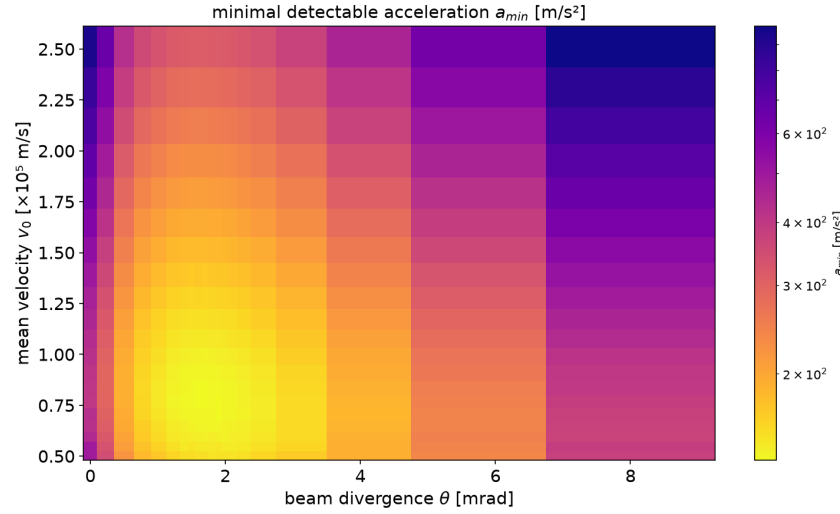


Figure 3.10: heatmap of the minimum detectable acceleration $a_{\min}$ of the Mach–Zehnder interferometer as a function of beam divergence $\theta_{\max}$ and mean velocity v_0 . The colour scale indicates $a_{\min}$ in $\frac{\text{m}}{\text{s}^2}$.

To compare the simulations with the literature estimates from [4], the estimated values must be rescaled to the same total number of shots as in the simulation. The simulations used $N_b = 400$ shots per scan point with $N_{\text{off}} = 200$ scan points, whereas the values in [4] correspond to approximately 400 shots of positronium atoms in total, for an entire scan. Since $a_{\min} \propto 1/\sqrt{N}$, the literature values are divided by $\sqrt{N_{\text{off}}} = \sqrt{200} \approx 14.1$.

Table 3.1 is not a point-by-point check of identical calculations. The two calculations summarized in the table have different statistics and different treatments of several important quantities, including the angular acceptance, fringe contrast, and the velocity distribution. The remaining differences in physical assumptions are retained and discussed below.

After rescaling the literature values to the simulation statistics of total number of shots $N_b N_{\text{off}} = 8 \times 10^4$, the moiré deflectometer result agrees with literature to within $\sim 25\%$, while for the Mach–Zehnder interferometer the simulation predicts a sensitivity about two times worse than [4], i.e. the simulated $a_{\min}$ is a factor of ~ 2 larger than the rescaled literature value. The remaining factor of about

Table 3.1: Comparison of the minimum detectable accelerations $a_{\min}$ obtained in the simulations and reported in the literature for a mean initial velocity of $v_0 = 10^5 \frac{\text{m}}{\text{s}}$. The simulations use 400 shots per scan point, while the literature estimates correspond to approximately 400 shots (Table 2 in [4]). The beam divergences used in each case are listed in the table. Here, $g = 9.81 \frac{\text{m}}{\text{s}^2}$.

Source	$\theta_{\max}$ [mrad]	$a_{\min}$ [$\frac{\text{m}}{\text{s}^2}$]	$a_{\min}/g$
moiré			
Simulation	17	3408 ± 31	347.4 ± 3.2
Literature	17	$\sim 4.25 \times 10^3$	$\sim 4.3 \times 10^2$
Mach–Zehnder			
Simulation	1.6	140 ± 1	14.27 ± 0.10
Literature	5.5	$\sim 6.9 \times 10^1$	~ 7.1

two could originate from the different treatment of the angular acceptance, fringe contrast and the velocity averaging. Since they are varied simultaneously, their individual effects cannot be separated here; doing so would be useful in future work. However, both the simulations and the literature estimates confirm the advantage of Mach–Zehnder interferometry over the moiré deflectometry for the same measurement time.



Conclusions

The main objective of this thesis has been to compare the sensitivity of two different inertial-sensing schemes, i.e. a classical moiré deflectometer and a Mach–Zehnder interferometer, by means of Monte Carlo simulation. For this purpose both devices have been modelled at single-atom level. Thus the atoms were propagated with velocities chosen from distribution based on measured 2^3S beam characteristics, a given angular spread and a lifetime chosen exponentially from an appropriate distribution with mean-lifetime of 1142 ns. The gravitational acceleration has then been reconstructed from the displacement (deflectometer) or the fitted phase (interferometer) of the transmission pattern [Eqs. (3.7) and (3.8)]. The reconstructed value has been compared with the input value and found to be unbiased within the given statistics for a value of applied acceleration of $100g$ (Fig. 3.8).

For a mean velocity of $v_0 = 10^5$ m/s and 3.2×10^6 s $\approx$ 5.3 weeks of acquisition time we find for the Mach–Zehnder interferometer a minimum detectable acceleration of (140 ± 1) m/s² ($14.27 \pm 0.10 g$) at a beam divergence of $\theta_{\max} \approx 1.6$ mrad (Figure 3.9). For comparison, at a beam divergence of $\theta_{\max} = 17$ mrad, adopted from Ref. [4] the moiré deflectometer yields $a_{\min} = (3408 \pm 31) \frac{\text{m}}{\text{s}^2}$, corresponding to $(347.4 \pm 3.2)g$ (Figure 3.4 and Table 3.1). Hence the interferometer is about 24 times more sensitive than the moiré deflectometer, in order-of-magnitude agreement with the analytical estimate of Ref. [4], which gives a ratio of several tens. The required number of atoms for a given target acceleration a scales as $a_{\min}^{-2}$. Thus the measurement time for the Mach–Zehnder interferometer is about 593

times shorter than for the moiré deflectometer.

An important distinction between the two instruments is how the sensitivity of the instrument depends on the beam. For the moiré deflectometer, the minimum detectable acceleration decreases monotonically with divergence (Fig. 3.4). The maximum number of atoms that pass through the collimation slit increases linearly with the collimation angle $\theta_{\max}$. The limiting factor is then the angular acceptance of the two gratings. For the Mach–Zehnder interferometer, there is an optimum value for the maximum divergence of the atoms, $\theta_{\max} \approx 1.6$ mrad (Fig. 3.9). For larger accepted angular divergences, the number of positronium atoms entering the interferometer increases approximately linearly in the present beam model. However, the reconstructed fringe visibility decreases with increasing divergence because atoms at larger incidence angles contribute less effectively to the interferometric signal due to the finite angular acceptance of the optical gratings. This is expressed in Eq. (3.9) [26]. The two-dimensional plot of the sensitivity for the interferometer (Fig. 3.10) clearly shows that this optimum is shallow. That is, near the minimum, the interferometer has a tolerance of several tens of percent in both velocity and divergence without a significant loss in sensitivity. This is an important property for a real apparatus, as it means that there is some flexibility in the preparation of the beam.

The sensitivities obtained in the present simulations are compared with the analytical estimates of Ref. [4] in Table 3.1, normalised to the same total number of shots. For the moiré deflectometer the simulation reproduces the analytical estimate to within $\sim 25\%$. For the Mach–Zehnder interferometer the simulated $a_{\min}$ is about a factor of 2 larger, i.e. the simulation predicts a sensitivity about two times worse than the estimate in Ref. [4]. The remaining difference originates from the different angular acceptance, fringe contrast and velocity averaging in the two calculations. Since, the above-mentioned assumptions are all different from each other, a quantitative separation of their individual contributions to the discrepancy is not possible within the scope of this work. The beam divergence of the interferometer has been estimated above as 1.6 mrad, whereas for the same configuration Ref. [4] stated a divergence of 5.5 mrad. The remaining difference

between the two estimates is due to normalization of the above counts to the number of atoms in the interferometer. When the conventions of Ref. [4] are applied consistently to both devices, the estimates agree, and the interferometer-to-deflectometer ratio is preserved, since a common interrogation time cancels in the ratio.

The interferometer result in the figure should be viewed as a statistical readout model of the interferometer rather than a full first-principles simulation of the interferometer of coherent positronium and laser. Thus, the sign and absolute value of $a_{\min}$ depend on the contrast and Bragg-acceptance parametrizations used in the model.

Two limitations should be stated here. First, the Mach–Zehnder interferometer is an intrinsically quantum device. A Geant4 cross-check was therefore not performed for the interferometer; for the classical deflectometer such a cross-check remains a useful validation and is left for future work. Second, the simulation does not include experimental systematic effects. The dominant ones for this scheme are the alignment and rotation of the gratings, the finite grating–stopper distance, and the pick-off annihilation of 2^3S atoms on the material elements, which affect the assignment of the annihilation position and the extracted contrast [4, 31]. The low polarizability of the 2^3S state limits its sensitivity to stray electric fields, which is one reason it is preferred over Rydberg states [3, 4].

These results support the feasibility of further studies toward a first inertial measurement on a purely leptonic matter–antimatter system. No direct gravitational measurement on positronium yet exists [1, 4]. The corresponding measurement on antihydrogen has only recently been reported [32]. The moiré technique is under development for antihydrogen within AEgIS [21]. For weak accelerations, the interferometer is preferred because its standing-wave gratings are almost fully transparent, so few atoms are lost, and its period is short [4, 25]. Reaching an acceleration of order $10g$ would require only a modest increase in the number of shots, compared to the simulated statistics. The gravitational regime relevant to a test of the weak equivalence principle needs far more integration time, or a larger usable flux. Ref. [4] identifies two routes to the latter: two-dimensional laser

Doppler cooling to reduce the beam divergence, and coherent Raman excitation of the 2^3S level to raise its production. The broad optimum found here shows that these steps need no stringent re-optimization of the beam. With the position-sensitive detection under development for this purpose [33, 17], a first test of the free fall of positronium becomes a realistic longer-term goal.

Declaration of artificial intelligence and information technologies usage in the process of thesis writing

I declare the usage of the following generative AI tools in thesis writing and preparation:

- Deepseek V3, V4-flash & R1: code generation and correction, literature checks, English grammar and vocabulary help, calculations verification
- Claude Sonnet 4.6: code generation
- Gemini 3.1 Pro, 3.5 Flash & 3.6 Flash: code correction, literature search
- Prism OpenAI Codex GPT-5.2: $\LaTeX$ and TikZ generation and correction, English grammar help
- Duck.ai gpt-5.0, gpt-5.4-nano & gpt-5.6-luna: answering my stupid programming questions

I wrote the thesis text myself, using AI tools for language correction and editorial assistance. No AI was used to generate the ideas or concepts of the thesis. After the usage of the listed tools I have verified and edited the content, and I take full responsibility for this publication.

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